Do Transformers Add Anything to a 3D CNN for Alzheimer's Disease Detection? A Leakage-Free, Statistically Controlled Benchmark on Structural MRI

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***Abstract*—Deep-learning systems for Alzheimer's disease (AD) detection from MRI routinely report 95–99% accuracy, and CNN–transformer hybrids are increasingly presented as the next step. Most of these figures, however, come from slice-level splits that place images of the same patient in both training and test sets, and hybrids are rarely compared against matched single-model baselines. We ask whether a transformer branch improves a 3D CNN when the evaluation is honest. Six fully 3D models — DenseNet-121, MedicalNet ResNet-18, ViT, Swin, and two late-fusion hybrids (CNN–ViT and Swin–ViT) — were trained on 1,132 T1-weighted volumes from 395 ADNI subjects with one preprocessing pipeline, one classifier head, and identical subject-level five-fold splits, and were compared with fold-paired tests, subject-level bootstrap intervals, and an age-and-sex-only baseline. On CN vs. AD, the CNN–ViT hybrid reached a balanced accuracy of 0.806 (95% CI 0.757–0.852), statistically indistinguishable from DenseNet alone (0.800; paired $p = 0.54$) despite twice the parameters. Replacing the CNN branch with Swin, by contrast, lowered macro-F1 in all five folds (balanced accuracy 0.670). On CN vs. MCI vs. AD, the best balanced accuracy was 0.549 against a chance level of 0.333, with MCI as the bottleneck, and a from-scratch ViT never predicted the majority class. The age-and-sex-only baseline stayed at chance, ruling out demographic shortcuts. 3D attribution maps disagreed across methods and partly highlighted non-brain tissue. At the scale of a typical ADNI study, the convolutional branch carries the signal and the transformer adds no measurable value.**

***Index Terms*—Alzheimer's disease, structural MRI, 3D CNN, vision transformer, hybrid models, data leakage, benchmark, explainability**

I. Introduction

Alzheimer's disease (AD) is the leading cause of dementia, and the atrophy it causes in the hippocampus and medial temporal lobe is visible on structural MRI [1], [2]. This has made MRI-based AD classification one of the most popular applications of deep learning in medical imaging [3].

The reported performance in this field depends heavily on how the data are split (Fig. 1). Studies that classify 2D slices or curated 2D images report a median accuracy of 99.0% [4]–[11]. Studies that keep every subject entirely in either the training or the test set report 60–90% [12], [13]. The difference is largely an artefact: when Yagis *et al.* [14] kept the data and model fixed and changed only the split from slice level to subject level, accuracy fell by about 28 points. Wen *et al.* [12] found possible leakage in more than half of the CNN studies they reviewed, a recent scoping review identifies leakage as the most recurrent source of performance inflation in AD deep learning [15], and even 3D CNNs can leak through longitudinal scans of the same subject [16].

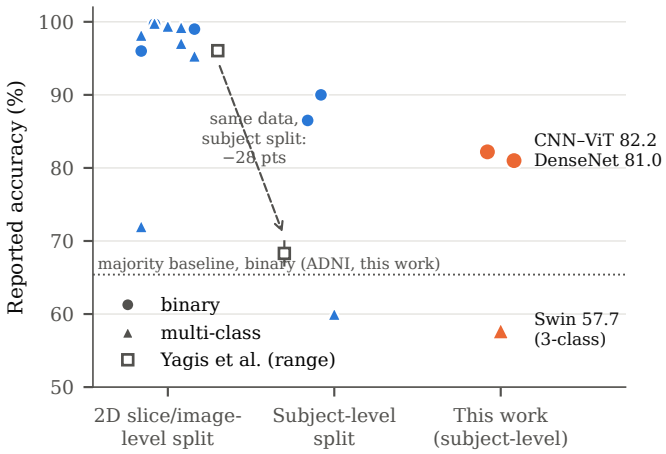


Fig. 1. **The leakage gap.** Accuracies reported in the literature, grouped by the level at which data were split. Squares show Yagis *et al.* [14], who changed only the split on the same data and model. Our subject-level results (orange) fall in the honest regime. Reported ranges are plotted at their midpoint. Dotted line: majority-class baseline of our binary task.

A second, less discussed problem concerns architecture claims. CNN–transformer hybrids are motivated by the intuition that convolutions capture local atrophy while self-attention captures long-range structure [5], [10]. Hybrids, however, are usually evaluated against weaker or differently tuned baselines, on different splits, and without statistical tests, so the contribution of the fusion itself is unknown. Under leakage-free evaluation it is not clear whether any of the gain survives.

We therefore design a controlled benchmark in which the data, subject-level splits, preprocessing, classifier head, imbalance handling, and evaluation are fixed across models, and use it to answer three questions:

- **RQ1:** Does adding a global transformer branch to a 3D CNN improve AD classification?
- **RQ2:** Which component carries the signal — the convolutional branch, the transformer, or pretraining?
- **RQ3:** Do 3D attribution maps of such models point to disease-related anatomy?

Our contributions are:

1) A leakage-free benchmark of six 3D models on ADNI with identical subject-level splits, preprocessing, and classifier head, reported with majority-class and demographic baselines, fold-paired tests, subject-level bootstrap confidence intervals, and model complexity.
2) A branch ablation showing that adding a ViT to a 3D CNN doubles the parameter count (11.2 M to 23.6 M) without a measurable gain ($p = 0.54$), whereas removing the CNN branch costs 0.14 macro-F1 in every fold.
3) Two documented failure modes: instability of a medically pretrained ResNet, and collapse of a from-scratch 3D ViT that never predicts the majority class.
4) A multi-method 3D attribution analysis that exposes method disagreement and non-brain attribution, together with a practical checklist for evaluating future hybrid models.

II. Related Work

Slice-level studies. ViT, ViT–GRU, multi-scale CNN, and CNN–Swin models trained on 2D slices report 95–99.8% accuracy [4]–[6], [10]. These studies use slice-level splits or curated 2D image sets, so their figures cannot be compared with subject-level results [14], [15].

Subject-level 3D studies. Wen *et al.* [12] reported 85–88% for CN vs. AD with 3D CNNs in a reproducible framework, and Bäckström *et al.* [17] showed that a subject-level split is essential for 3D CNNs. Saoud and AlMarzouqi [13] used one baseline scan per subject and an ensemble of 138 regional 3D ViTs, reporting 90% for CN vs. AD and 60% for the three-class task. A concurrent study reports 72.3% with a multi-modal 3D CNN on OASIS-1 under subject-level cross-validation [18]. At CVPR, M3T [19] combined 3D and 2D CNNs with a transformer, trained on 4,786 ADNI scans, and tested on independent ADNI, AIBL, and OASIS sets. It reached 93.2% on ADNI, 1.2 points above a 3D DenseNet-201, but from a single run without significance tests. None of these works isolates the contribution of a transformer branch inside a hybrid; that is the gap we address.

III. Benchmark Design

A. Leakage-Free Protocol

Let $\mathcal{D} = \{(x_i, y_i, s_i)\}_{i=1}^{N}$ be the scans, where x_i is a 3D volume, y_i its label, and s_i its subject. We partition the set of *subjects* into five stratified folds $S_1, \ldots, S_5$ (seed 42) and test on fold k after training on the remaining folds, so that

$$\{s_i : i \in \text{train}_k\} \cap \{s_i : i \in \text{test}_k\} = \varnothing. \tag{1}$$

Inside each training partition, 12% of the subjects are held out at random (seed $42 + k$) as a validation set for early stopping and checkpoint selection, so the test fold is never used for model selection. Because subjects contribute 2.9 scans on average, the softmax probabilities of a subject's scans are averaged:

$$\hat{y}_s = \arg\max_c \frac{1}{|I_s|} \sum_{i \in I_s} p_\theta(c \mid x_i), \qquad I_s = \{i : s_i = s\}. \tag{2}$$

All metrics are computed at subject level. We report balanced accuracy (BAcc, the mean per-class recall), macro-F1, accuracy, and, for the binary task, sensitivity and specificity with AD as the positive class. The reference points are the majority-class baseline (always CN: accuracy 65.4% binary and 52.2% three-class) and chance BAcc (0.5 and 0.333). Accuracy and macro-F1 are reported as mean $\pm$ SD over folds. BAcc is reported with a 95% subject-level bootstrap confidence interval (10,000 resamples of the pooled out-of-fold predictions). Models are compared with a paired t-test on fold-wise macro-F1, and we also report how many folds each model wins. To test for demographic shortcuts, we also train a class-balanced logistic regression on age and sex alone, on the same folds.

B. Cohort and Preprocessing

We used ADNI [20], [21] 3T T1-weighted MPRAGE volumes (ADNI GO/2/3, pre-processed N3m NIfTI). After removing 18 off-protocol scans (hippocampal segmentations and AC–PC-registered variants), the cohort contains 206 CN (627 scans), 80 MCI (182), and 109 AD (323) subjects, with up to six scans per subject spanning at most two years (median one year). Every scan inherits its subject's ADNI research group, which is constant across each subject's scans. The groups are matched for age (CN 74.8 ± 7.1, MCI 73.9 ± 8.2, AD 74.7 ± 7.9 years; CN vs. AD $p = 0.93$) and sex (53%, 46%, and 44% female; χ^2 $p = 0.24$). A deterministic MONAI [22] pipeline reorients each volume to RAS, resamples it to 1.5 mm isotropic spacing, crops the foreground, applies z-score normalization over non-zero voxels, and pads or crops it to 128^3 (DenseNet and ResNet) or 96^3 (all transformer and hybrid models, whose patch embedding and memory footprint require it). No skull-stripping or registration is applied. Training-time augmentation uses left–right flips, random affine transforms ($\pm 10°$, $\pm 10\%$ scale, 8-voxel translation), Gaussian noise and smoothing, bias field, gamma (0.8–1.2), and intensity scaling and shifting.

C. Controlled Model Family

All backbones produce a feature vector f that feeds a shared head (dropout 0.2 and a linear layer). We test four single backbones and two hybrids:

- **DenseNet-121** [23]: 3D, trained from scratch, 1024-D features.
- **ResNet-18** initialized with MedicalNet weights [24]: 512-D.
- **ViT** [25]: 384-D, 6 layers, 6 heads, 16^3 patches, trained from scratch.
- **Swin**: the SwinUNETR encoder [26], [27] (feature size 48, four stages).

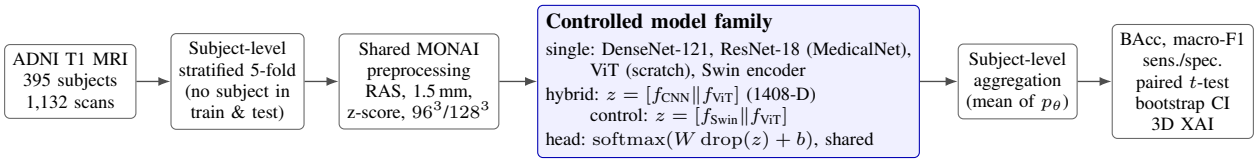


Fig. 2. **Benchmark design.** Every model sees the same subjects, splits, preprocessing, classifier head, imbalance handling, and evaluation; backbone-specific hyperparameters are listed in Table II. The Swin–ViT hybrid is an ablation control that removes the convolutional branch.

TABLE I

CN VS. AD (315 SUBJECTS). ACCURACY AND MACRO-F1: MEAN ± SD OVER FIVE SUBJECT-LEVEL FOLDS. BACC, SENSITIVITY, AND SPECIFICITY: POOLED OUT-OF-FOLD SUBJECT PREDICTIONS, WITH A 95% BOOTSTRAP CI. Δ: FOLD-PAIRED DIFFERENCE IN MACRO-F1 AGAINST THE REFERENCE MODEL, WITH THE FOLDS WON.

Model	Accuracy	Macro-F1	BAcc [95% CI]	Sens.	Spec.	Δ macro-F1 (paired p; folds won)
CNN–ViT hybrid	**0.822**±0.057	**0.802**±0.062	**0.806** [0.757, 0.852]	0.752	**0.859**	reference
DenseNet-121	0.810±0.041	0.788±0.062	0.800 [0.752, 0.846]	**0.771**	0.830	−0.014 ($p = 0.54$; 2/5)
Swin–ViT hybrid (no CNN)	0.702±0.070	0.662±0.101	0.670 [0.615, 0.724]	0.569	0.772	−0.140 ($p = 0.11$; 0/5)
ResNet-18 (MedicalNet)	0.698±0.139	0.654±0.168	0.689 [0.635, 0.742]	0.670	0.709	−0.148 vs. CNN–ViT
Age + sex only (logistic)	*0.498±0.092*	*0.486±0.092*	*0.503*‡	–	–	
Majority baseline (always CN)	*0.654*	*0.395*	*0.500*	*0.000*	*1.000*	

‡Mean over folds (the demographic baseline has no pooled predictions).

TABLE II

MODEL CONFIGURATIONS. PARAMETERS AND FORWARD GFLOPS (ONE VOLUME) ARE MEASURED WITH THE RELEASED CODE. EP/WU/PAT: MAXIMUM EPOCHS, WARMUP EPOCHS, EARLY-STOPPING PATIENCE. ALL MODELS USE ADAMW AND THE SHARED HEAD.

Model	Input	Par. (M)	GFLOPs	Ep/Wu/Pat	LR	WD
Binary (class-weighted CE, label smoothing 0.1)						
DenseNet-121	128^3	11.2	85.8	80/5/15	1e-4	0.01
ResNet-18 (MedicalNet)	128^3	33.0	501.6	60/5/12	1e-4[a]	0.01
CNN–ViT	96^3	23.6	41.9	100/8/18	1e-4	0.03
Swin–ViT	96^3	20.4[b]	55.5[b]	100/8/18	1e-4	0.03
Three-class (class-weighted focal loss, $\gamma = 2$)						
DenseNet-121	128^3	11.2	85.8	80/5/15	1e-4	0.01
Swin	96^3	8.1	49.9	100/5/18	5e-5	0.02
ViT	96^3	12.3	5.7	100/8/18	1e-4	0.05

[a]Backbone frozen for 5 epochs, then trained at 1e-5. [b]Concatenation variant in the released code.

- **CNN–ViT hybrid**: runs DenseNet and ViT on the same volume and concatenates their features, $z = [f_{\text{CNN}} \| f_{\text{ViT}}]$.
- **Swin–ViT hybrid**: Swin in place of DenseNet, so that comparing it with CNN–ViT indicates the contribution of the convolutional branch. This run used an earlier ViT-branch variant with an additional cross-attention block that is not in the released code, so the control is close to, but not exactly, the CNN–ViT design.

Fig. 2 summarizes the design, and Table II lists each model's input, size, cost, and schedule.

D. Training

All models are trained with AdamW [28], linear warmup followed by cosine decay, mixed precision, an effective batch size of 8 (2×4 gradient accumulation), and gradient clipping at 1.0, on one 16 GB RTX 5060 Ti. Binary models use class-weighted cross-entropy with label smoothing 0.1 [29]; three-class models use class-weighted focal loss [30] ($\gamma = 2$). Every run also draws training batches with a class-balanced sampler, so class imbalance is corrected twice (sampler and loss weights). Per-model schedules (Table II) were set by the developers for each backbone; the pretrained ResNet uses a frozen-backbone warm start and a $10\times$ lower backbone learning rate. The checkpoint with the best validation subject-level macro-F1 is evaluated once on the test fold.

E. Explainability Protocol

We computed Grad-CAM [31], Grad-CAM++ [32], Score-CAM, Integrated Gradients [33], SmoothGrad [34], occlusion, LIME [35], and attention rollout [36] in 3D, and inspected whether the methods agree with one another and with known AD anatomy.

IV. RESULTS

Finding 1 — Adding a ViT branch to a 3D CNN gives no measurable gain (RQ1). The CNN–ViT hybrid has the highest mean scores (Table I), but its advantage over DenseNet alone is $+0.014$ macro-F1. A paired t-test gives $t = 0.67$ with 4 degrees of freedom ($p = 0.54$); the hybrid wins three of five folds, and the BAcc confidence intervals of the two models almost coincide. Fig. 3a shows that the two models cross each other from fold to fold. Over all 315 subjects the hybrid classifies 259 correctly and DenseNet 255. The hybrid has $2.1\times$ as many parameters (23.6 M vs. 11.2 M), so the ViT branch adds capacity without a detectable benefit. Because the two configurations also differ in input size and schedule (Table II), this is a comparison of tuned models rather than a single-factor ablation; a matched-input DenseNet run would make it strict.

Finding 2 — The convolutional branch carries the signal; pretraining does not (RQ2). Removing the CNN branch

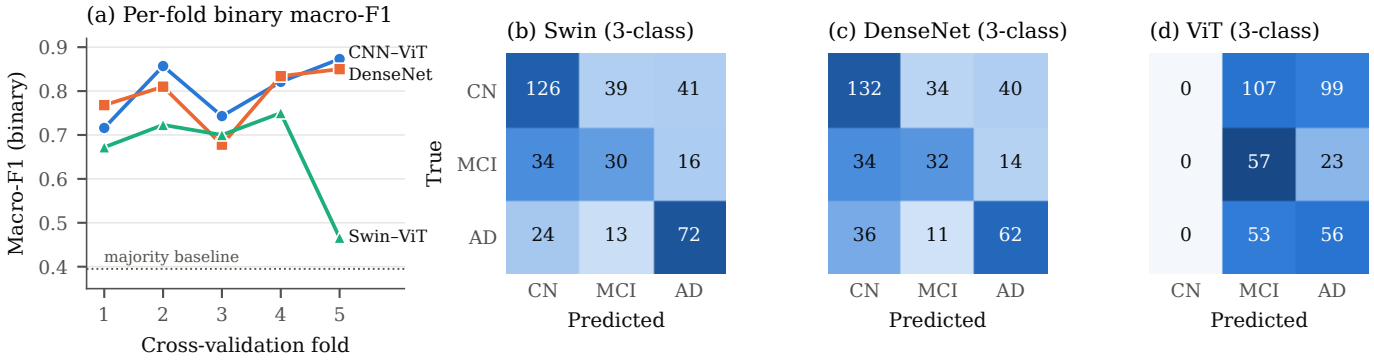


Fig. 3. **(a)** Per-fold binary macro-F1 on identical subject-level folds: CNN–ViT and DenseNet alternate, whereas Swin–ViT (no CNN branch) is lower in every fold. **(b–d)** Pooled three-class confusion matrices (subject counts; shading is row-normalized). MCI is mostly absorbed into CN and AD, and the from-scratch ViT never predicts CN.

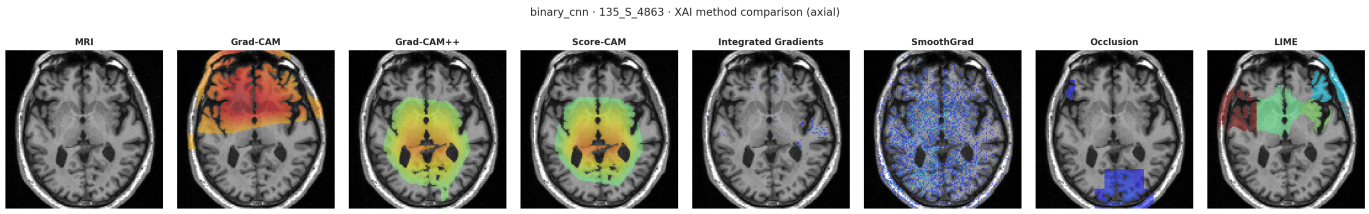


Fig. 4. **Attribution methods disagree.** Eight 3D attribution methods applied to the binary DenseNet on one AD subject (axial slice): Grad-CAM points frontally, Grad-CAM++/Score-CAM centrally, occlusion to the occipital region and skull, and LIME partly to the scalp.

TABLE III
CN VS. MCI VS. AD (395 SUBJECTS). CHANCE BACC IS 0.333.

Model	Macro-F1	BAcc [95% CI]	MCI recall
Swin	**0.537**±0.078	**0.549** [0.499, 0.601]	0.375
DenseNet-121	0.524±0.071	0.537 [0.485, 0.588]	**0.400**
ViT (scratch)	0.256±0.012	0.409 [0.363, 0.454]	0.713
Age + sex only	*0.371±0.018*	*0.380*‡	–
Majority baseline	*0.229*	*0.333*	*0.000*

(Swin–ViT) lowers macro-F1 by 0.14 and loses in all five folds. The paired t-test is not significant ($p = 0.11$) because of a single extreme fold, but the BAcc confidence intervals do not overlap (0.615–0.724 vs. 0.757–0.852), and Swin–ViT misses 43% of AD subjects. The MedicalNet-pretrained ResNet is only about four points above the majority baseline in accuracy and is bimodal across folds: three folds reach 0.78–0.82 macro-F1 and two fall to 0.449. Its schedule makes a late take-off likely: the backbone is frozen for five epochs and then trained at a 10× lower learning rate, while early stopping uses the shortest patience of all models (12 epochs). We have not yet re-run it with a minimum-epoch guard to confirm this. At this data scale, inductive bias from convolution matters more than either global attention or medical pretraining. The signal is also not demographic: the groups are age- and sex-matched, and an age-and-sex-only classifier on the same folds stays at chance (BAcc 0.50 binary, 0.38 three-class; AUC 0.48 and 0.54).

Finding 3 — Three-class classification is well above chance, but MCI is the bottleneck. Swin and DenseNet are tied (paired $p = 0.52$) at a BAcc of about 0.54–0.55, well above chance (Table III). Their per-class F1 is about 0.65 for CN and 0.55–0.61 for AD, but only 0.37–0.41 for MCI. Fig. 3b–c shows that MCI subjects are mostly absorbed into CN and AD, as expected for a clinically intermediate stage. The from-scratch ViT reaches 28.6% accuracy, below the 52.2% majority baseline, because it *never* predicts CN (Fig. 3d). Its high MCI recall is therefore an artefact of predicting MCI or AD for every subject, not evidence of skill. This pattern is consistent with a data-hungry model [25] that has learned nothing useful, pushed toward the minority classes by the double imbalance correction (class-balanced sampling plus class-weighted focal loss).

Finding 4 — Without skull-stripping, attribution maps are not anatomically conclusive (RQ3). On the same subject, the attribution methods highlight different regions (Fig. 4). For the Swin and Swin–ViT models, group-averaged Integrated Gradients over 16 AD subjects is nearly uniform across thirds of the volume (0.036–0.055), with mass on the cerebellum, orbits, and scalp rather than a medial-temporal focus. Similar identity and shortcut cues have been reported for 3D MRI models [16]. A single favourable heatmap would therefore be misleading, and attribution claims require skull-stripped inputs, atlas ROIs, and sanity checks.

V. Discussion

Our results fall in the same honest regime as other subject-level studies (Fig. 1): 82% binary accuracy, against 85–88% [12] and 90% [13], and 58% three-class accuracy against 60% [13]. We obtain these with a single model and without registration or skull-stripping, which plausibly explains the remaining gap. The central lesson concerns architecture claims. The baseline table of M3T [19] points the same way: with four times our training data, adding a transformer to DenseNet-201 *lowered* ADNI accuracy from 92.0% to 90.4%, and a pure 3D ViT was again the weakest model. At around 300–400 subjects, the benefit usually attributed to hybrids came from the convolutional branch, and it disappears when a hybrid is compared with an equally tuned CNN on the same splits.

Checklist for evaluating hybrid models. Based on these findings, we recommend:

1) Split at the subject level and aggregate scans per subject.
2) Report the majority-class, chance, and demographics-only baselines next to every score.
3) Compare a hybrid against each of its branches trained alone, on identical folds.
4) Test differences with fold-paired tests and confidence intervals, and report the number of folds won.
5) Report per-fold results, not only mean $\pm$ SD.
6) Skull-strip the inputs and use more than one attribution method before making anatomical claims.

Limitations.

- *Incomplete model grid.* Standalone ViT and Swin were not run on the binary task, and CNN–ViT was not run on the three-class task.
- *Statistical power.* Five folds and a single seed limit the power of the tests, so a small benefit of hybridization cannot be ruled out.
- *Not a single-factor ablation.* Input size and schedules differ between some models (Table II), and the Swin–ViT control used an earlier ViT-branch variant.
- *Short follow-up and enrolment labels.* Labels are the ADNI enrolment groups, applied to scans acquired within two years.
- *External validation.* The study uses only the ADNI cohort.

VI. Conclusion

Under leakage-free, statistically controlled evaluation, a transformer branch did not improve a strong 3D CNN for AD classification. The convolutional branch carried the signal, the three-class task was limited by MCI, and attribution maps on non-skull-stripped inputs were not anatomically conclusive. Hybrid-model claims should be backed by matched single-branch baselines and paired tests. Future work will complete the model grid with repeated cross-validation, add skull-stripping, atlas-ROI attribution, and an age-only baseline, and validate the models on independent cohorts.

Code Availability

Code, configurations, and the subject-level split files are available at https://github.com/SamuelCormaken/Deep-Learning-for-Alzheimers-MRI. ADNI images must be obtained from ADNI under its Data Use Agreement.

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